

tric worldview to one that is software-defined. This shift required reimagining everything from the operating system to the deployment of containers and microservices in various applications. The project demonstrated how to accelerate deployment cycles by leveraging cloud-native technology principles, thus bringing a new level of agility and adaptability to traditionally rigid industries.

One of the significant challenges addressed by Bobby Car was establishing a robust base that adhered to functional safety standards, beginning with the operating system. Extending this approach through the entire value chain, the project facilitated the development and testing of new features, ensuring seamless integration without disrupting existing systems. This methodology was crucial, especially for connected vehicle platforms and IoT architectures, where reliability and safety are paramount.

A key aspect of the Bobby Car project was its emphasis on learning from past experiences and best practices. This knowledge was crucial in developing a strategy that was not only innovative but also practical and sustainable. Continuous integration and deployment became fundamental, as was the ability to conduct virtual testing and create digital twins to enhance efficiency and provide improved solutions to customers.

A pivotal element of the Bobby Car initiative was also its integration with a broader ecosystem, encompassing everything from payment systems to 5G technology and the manufacturing sector. This comprehensive approach ensured that applications developed under the Bobby Car project could seamlessly interact with various components of the industry, enhancing the user experience and operational efficiency.

Key components and applications

Bobby Car consists of two primary components: the “vehicles waiter” and the “geography console.” The vehicles waiter is an intricate system providing detailed insights into various metrics such as delivery data, temperature, speed, RPM, and more, leveraging data from the OBD port of vehicles. This feature is crucial for real-time data analysis and decision-making.

The geography console, on the other hand, is implemented as a series of Kubernetes clusters. This setup enables the deployment of CRD objects, facilitating the ability to introduce isolated objects within the software stack. This aspect is particularly beneficial for conducting over-the-air (OTA) updates, allowing for modifications without disrupting other parts of the system.

One of the most significant advantages of Bobby Car is its data handling capability. The project utilises a DevSecOps pipeline, ensuring that data application and deployment are both efficient and secure. This pipeline



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